

Build-up and dephasing of Floquet–Bloch bands on subcycle timescales

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Strong light fields have created opportunities to tailor novel functionalities of solids^{1–5}. Floquet–Bloch states can form under periodic driving of electrons and enable exotic quantum phases^{6–15}. On subcycle timescales, lightwaves can simultaneously drive intraband currents^{16–29} and interband transitions^{18,19,30,31}, which enable high-harmonic generation^{16,18,19,21,22,25,28–30} and pave the way towards ultrafast electronics. Yet, the interplay of intraband and interband excitations and their relation to Floquet physics have been key open questions as dynamical aspects of Floquet states have remained elusive. Here we provide this link by visualizing the ultrafast build-up of Floquet–Bloch bands with time-resolved and angle-resolved photoemission spectroscopy. We drive surface states on a topological insulator^{32,33} with mid-infrared fields—strong enough for high-harmonic generation—and directly monitor the transient band structure with subcycle time resolution. Starting with strong intraband currents, we observe how Floquet sidebands emerge within a single optical cycle; intraband acceleration simultaneously proceeds in multiple sidebands until high-energy electrons scatter into bulk states and dissipation destroys the Floquet bands. Quantum non-equilibrium calculations explain the simultaneous occurrence of Floquet states with intraband and interband dynamics. Our joint experiment and theory study provides a direct time-domain view of Floquet physics and explores the fundamental frontiers of ultrafast band-structure engineering.

In a crystalline solid, the discrete translational symmetry of the ionic potential implies that the band structure of Bloch electrons is periodic in momentum space. Analogously, a time-periodic potential of light with frequency Ω acting on a Bloch electron replicates its band structure as ‘Floquet’ copies^{1,4,5} shifted in energy by multiples of $\hbar\Omega$, where $\hbar$ is the reduced Planck constant. When the lightwave potential approaches the strength of the static crystalline one, Floquet–Bloch sidebands hybridize among each other^{1,6–10} and with ground-state bands¹¹, shaping a quasi-stationary band manifold under periodic driving (Fig. 1a). Floquet engineering has raised high hopes to create novel quantum effects, such as dynamical localization⁵, photon-dressed topological states^{6–10} and time crystals^{12,13}. Besides proof-of-concept experiments in atomic systems^{12,13}, quasi-stationary Floquet–Bloch states in topological insulators have been studied^{7,8}, and emergent quantum phenomena such as a light-induced anomalous Hall effect in graphene¹⁴ and magnonic space–time crystals have been observed¹⁵. Yet, the dynamics of Floquet states, such as their transient emergence and the transfer of electrons between sidebands^{34,35}, has not been experimentally accessible.

Meanwhile, the interaction of atomically strong light fields with Bloch electrons has been explored on timescales shorter than a single cycle of light^{18–24,26–31}. The carrier wave has been shown to ballistically accelerate electrons through individual bands^{16–29} and to simultaneously induce

non-perturbative multiphoton interband transitions^{18,19,30} (Fig. 1a). The resulting dynamics have been utilized for high-harmonic generation (HHG)^{16,18,19,21,22,25,28–30}, electron–hole recollisions^{17,20,27}, lightwave valleytronics²⁶ or carrier-envelope-phase-controlled photocurrents²³. Although intraband currents driven by relatively weak terahertz fields have been visualized in angle-resolved photoemission spectroscopy (ARPES)²⁴, the description of strong-field interband excitations and their interplay with intraband dynamics has relied on theoretical models. More generally, the connection between subcycle dynamics and Floquet states associated with multicycle periodic driving has remained largely elusive despite its crucial role in strong-field light–matter interaction and its potential for shaping dynamic phases of matter by light.

Here we directly visualize the ultrafast build-up and dephasing of a Floquet–Bloch band structure with subcycle ARPES. The first half-cycle of an intense mid-infrared (MIR) lightwave accelerates electrons within the Bloch band structure before Floquet sidebands emerge within a single optical cycle. Simultaneously, intraband acceleration proceeds within multiple sidebands, until high-energy electrons scatter into bulk states and dissipation destroys the Floquet bands. Our lightwave ARPES observations are in line with quantum non-equilibrium calculations and establish the ultrafast limit of optical band-structure design.

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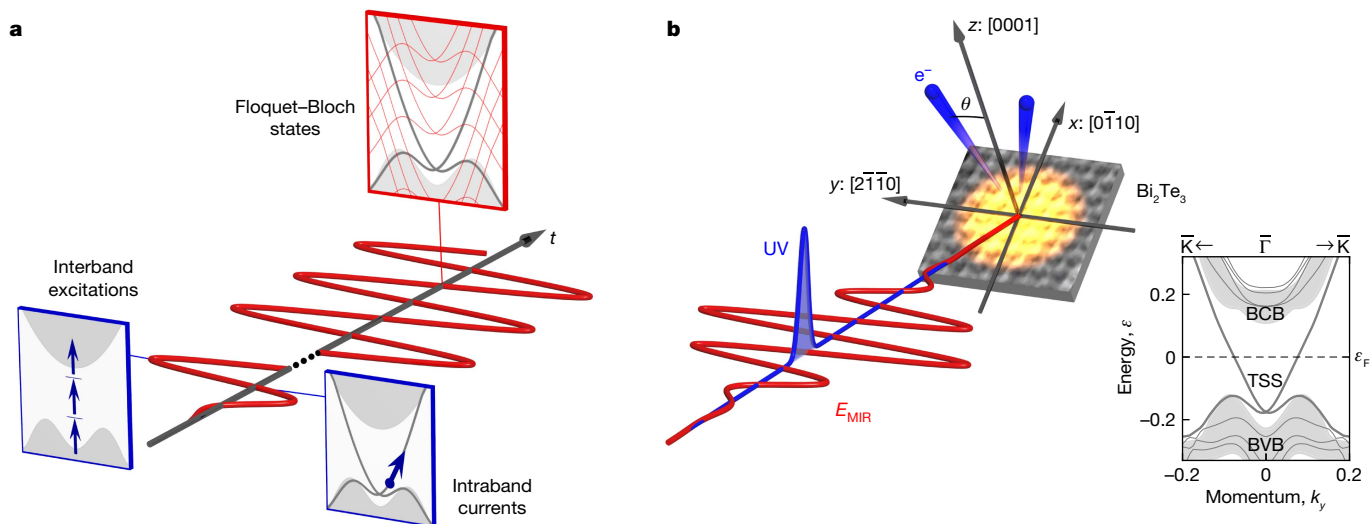


Fig. 1 | Conceptual idea of Floquet band engineering on the subcycle scale. **a**, Illustration of fundamental electronic processes induced by intense MIR light fields in solids. Whereas multiphoton interband excitation as well as lightwave-induced intraband currents are expected to occur already in the subcycle regime, strong-field periodic driving may result in Floquet–Bloch states. Yet the transition between these regimes is unclear. **b**, Schematic of the experiment: an s-polarized, phase-stable MIR electric field E_{MIR} (red waveform)

excites the topological surface state (TSS) of Bi_2Te_3 . ARPES spectra recorded with p-polarized UV probe pulses (blue) at variable time delay provide snapshots of the induced dynamics in the electronic band structure $\epsilon(k_y)$ with subcycle temporal resolution. Here, θ is the photoelectron emission angle and ϵ_F is the Fermi level. The direct mapping of the electronic band structure includes the bulk valence band (BVB) and the bulk conduction band (BCB).

Strong-field band-structure videography

The three-dimensional topological insulator bismuth telluride (Bi_2Te_3) forms an ideal platform for our study. This material features a finite bandgap in the bulk and a Dirac-like topological surface state (TSS) (Fig. 1b). Spin-momentum locking^{32,33} protects the TSS from scattering²⁴, which warrants sufficient coherence times for Floquet–Bloch states to emerge under periodic driving^{7,8,36}. We excite the system with intense, phase-locked, s-polarized MIR pulses with a centre frequency of $\Omega_{\text{MIR}}/2\pi = 25$ THz (photon energy, $\hbar\Omega_{\text{MIR}} = 0.1$ eV) (Fig. 1b). Simultaneously, the ultrafast evolution of the band structure and its occupation are mapped out by lightwave ARPES: an ultraviolet (UV) probe pulse arriving at a variable delay time, t , photoemits electrons by a two-photon process; the energy and momentum of the photoelectrons are detected with a hemispherical electron analyser. For a time resolution better than half an oscillation period of the MIR driving field, we compress the photoemission probe to 17 fs (Extended Data Fig. 1), which inevitably leads to spectral broadening of the ARPES data. The visibility of fine band-structure details on the energy scale of $\hbar\Omega_{\text{MIR}}$ can be augmented by second-derivative image processing, which is well established in high-resolution ARPES³⁷ to precisely reconstruct the original peak positions with the desired sensitivity (Extended Data Fig. 2).

The intense MIR field not only drives dynamics inside the sample but also interacts with emitted photoelectrons in vacuo²⁴. P-polarized MIR light induces well defined displacements of the whole projected band map along the energy axis in a process called energy streaking²⁴, whereas s-polarized light shakes the momentum coordinates. Such momentum streaking allows for a quantitative in situ reconstruction of the electric-field waveform directly on the sample surface, which is difficult to access otherwise²⁴. The procedure works well also for curvature-filtered ARPES spectra (Extended Data Fig. 3). The valence-band streaking trace (Fig. 2a) is readily converted into an s-polarized electric-field waveform (Fig. 2b), evidencing the subcycle resolution of our set-up at MIR frequencies. The retrieved peak field of 0.8 MV cm^{-1} corresponds to an incident amplitude of about 7 MV cm^{-1} (Methods) and is strong enough for HHG in the TSS²⁸.

Build-up of Floquet–Bloch sidebands

Streaking-compensated ARPES maps²⁴ reveal the material-inherent dynamics of the band structure (Fig. 2c). Up to 100 fs before the maximum field crest, the effect of the MIR lightwave on the TSS is negligible. The ARPES image shows a partially filled Dirac cone, in agreement with our density functional theory (DFT) calculations for Bi_2Te_3 . Forty femtoseconds later ($t = -60$ fs), the electric field has induced a strong asymmetry in the electronic distribution along the momentum axis—the hallmark of lightwave-driven currents. Most remarkably, a distinct splitting of electronic populations into multiple branches becomes noticeable only one-quarter of an oscillation cycle later, at $t = -50$ fs. Simultaneously, the electron population oscillates back and forth along the momentum axis while the driving field evolves (Fig. 2c, $t = -50$ fs and $t = -40$ fs). Figure 2b (red circles) shows the temporal evolution of the spectrally integrated current j_y retrieved by curvature filtering with less peak sensitivity (Extended Data Fig. 3). Even under the extreme conditions of our strong-field MIR lightwaves, the total subcycle current is well described by the semiclassical Boltzmann equation (Fig. 2b, red curve), treating electrons like Dirac particles. Yet, the emergent band splitting indicates quantum coherences beyond any semiclassical model.

The split bands notably follow the ground-state band structure shifted by integer multiples of $\hbar\Omega_{\text{MIR}} = 0.1$ eV (Fig. 2c, thin grey curves). The visibility of the spectral modulations is less pronounced at lower fields with all other features remaining intact (Extended Data Fig. 4). As the field increases, the band splitting becomes more prominent and the probability to find the electrons shifts entirely to the upper branch near the field crest (Fig. 2c, $t = -18$ fs), indicating the signatures of Floquet–Bloch states. This observation is surprising because the conventional Floquet picture assumes multicycle lightwaves to form sidebands by coherent superposition of electronic states^{7,8}, whereas our subcycle experiments identify Floquet–Bloch states already within the first two cycles of an MIR pulse. We can exclude alternative effects to cause these features: potential artefacts of the curvature analysis can be ruled out because the band splitting is well resolved in the raw ARPES spectra (Extended Data Figs. 5 and 6).

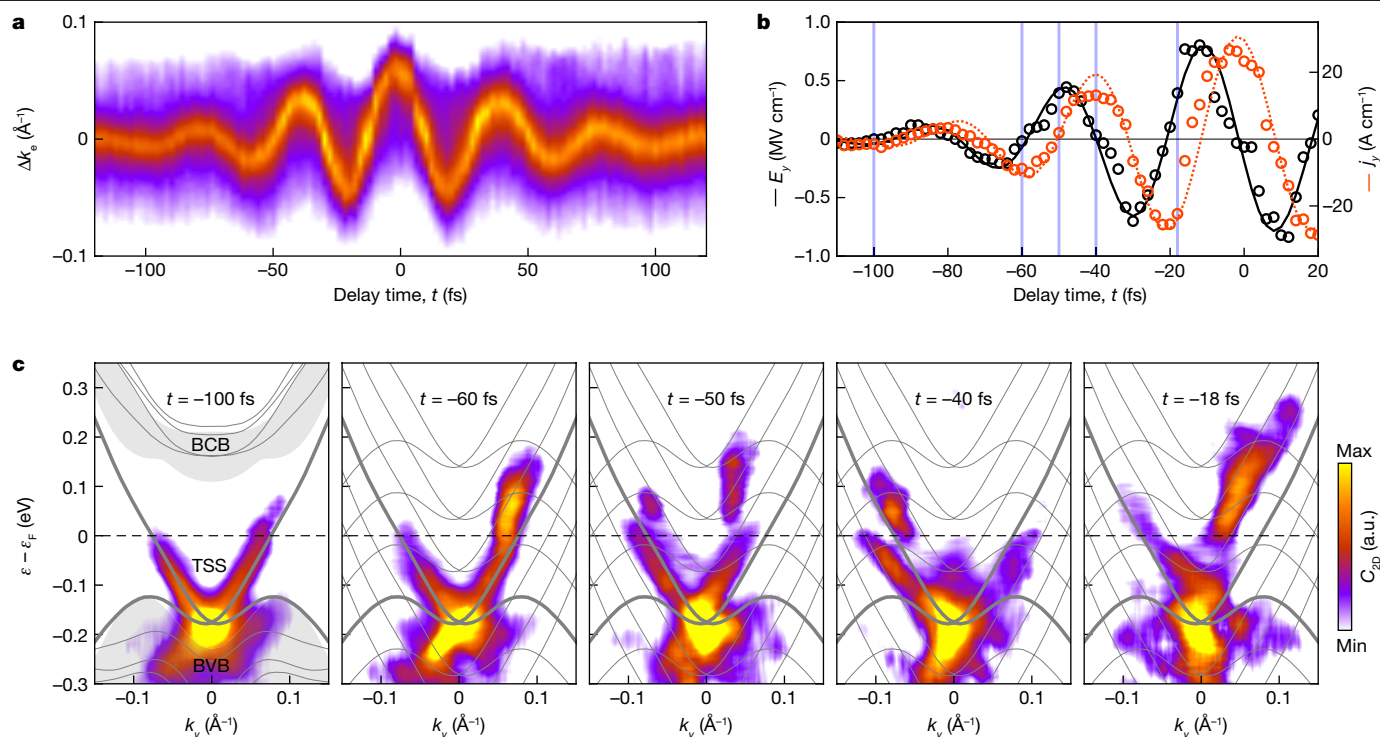


Fig. 2 | Subcycle observation of Floquet–Bloch sidebands. **a**, Curvature-filtered photoelectron momentum-streaking trace of the 25-THz driving field. Δk_x denotes the momentum deflection of photoelectrons, and the colour scale indicates photoelectron intensity processed by the two-dimensional curvature filter (Methods), C_{2D} . **b**, First half of the electric-field waveform E_y reconstructed from **a** (black circles) and current density j_y extracted from the measured asymmetry in the population of the surface bands (red circles). The black solid curve is a fitted analytic function of the electric field with a peak amplitude of 0.8 MV cm^{-1} and the dotted red line is the electric current induced by this waveform according to the semiclassical Boltzmann model. Thin vertical lines

indicate the temporal positions of the ARPES maps shown in **c**. **c**, Curvature-filtered lightwave ARPES maps recorded before the arrival of the MIR field ($t = -100 \text{ fs}$) and at selected delay times during the first half of the driving pulse, at $t = -60 \text{ fs}$, $t = -50 \text{ fs}$, $t = -40 \text{ fs}$ and $t = -18 \text{ fs}$. The images are compensated for the streaking shifts in **a**. The DFT band structure is superimposed in the leftmost panel. The electronic distributions are initially accelerated along the ground-state band dispersion, but subsequently divert from it, splitting into multiple branches that follow approximately the TSS band structure shifted by integer multiples of the driving frequency (thin grey curves).

So-called laser-assisted photoemission (LAPE)³⁸ or Volkov states^{8,36} created by the interaction of the emitted photoelectrons with the MIR lightwave have been shown to be strongly suppressed when the driving field is polarized in the surface plane⁸, which is our configuration (Fig. 1b). Indeed, lightwave ARPES maps recorded for a non-zero MIR electric-field component perpendicular to the surface differ strongly from Fig. 2c (Extended Data Fig. 7). We also verified that the two-photon probing is performed under non-resonant conditions and thus cannot introduce sideband artefacts (Extended Data Fig. 8). It is noted that hybridization between Floquet–Bloch sidebands is known to modify their dispersion and open small gaps⁷⁸. The relevant dispersion direction, however, is perpendicular to the polarization vector of the driving electric field and not parallel as in our experimental set-up.

To understand how MIR-driven intraband acceleration leads to the emergence of Floquet sidebands already in the second oscillation cycle of the carrier field (Fig. 3a), we track the spectrogram of a single Dirac electron (Fig. 3b) with a minimal quantum model (Methods). The driving MIR field periodically modulates the phase of the electron wave function $\Psi(t)$. After one oscillation cycle, constructive superposition of amplitudes with the same phase becomes possible and leads to the appearance of peaks in the spectral density $|\Psi(\varepsilon)|^2$ located at the energies $\varepsilon = \pm \hbar\Omega_{\text{MIR}}$. In the absence of dephasing, these Floquet sidebands continuously sharpen as the number of cycles increases. Sidebands of order n gain relevant weight only if the driving field accelerates electrons to a quiver energy (Fig. 3b, black curve) of $n \times \hbar\Omega_{\text{MIR}}$ (Methods).

For a more detailed microscopic description, we perform full quantum simulations of the subcycle-ARPES spectra with MIR driving based on a non-equilibrium Green's functions approach (Methods). The resulting spectra (Fig. 3c and Extended Data Fig. 9) reproduce the salient experimental features very well: starting from the ground-state distribution, the electrons are accelerated, initially within the original Bloch band. At early times, quantum interference is not strong enough to cause band splitting but becomes already visible in terms of an apparent steepening of the band ($t = -60 \text{ fs}$, Fig. 3c). Clearly discernible sideband splitting sets in near the second MIR field crest ($t = -50 \text{ fs}$) with the intensity distributed nearly equally between both branches. A quarter of an optical cycle later, the split intensity has shifted almost completely to the left ($t = -40 \text{ fs}$). A further half-cycle later ($t = -20 \text{ fs}$), when most of the electrons have been accelerated to the right branch, the MIR field has also increased and most of the intensity now appears in the first Floquet–Bloch sideband.

On the basis of these calculations, one can consistently understand both intraband currents and the emergence of Floquet–Bloch bands from the micromotion of electrons subject to a strong driving field. Essentially, the Floquet states are the result of a rapidly growing phase space for quantum trajectories upon strong driving and their constructive and destructive interference. Importantly, the averaged electron momenta follow the classical expectation over many cycles although the trajectory of a single electron is well defined only in the subcycle regime (Fig. 3b, black curve). While the electrons move through the band structure they collect quantum phases, which define the shape and the weight of the Floquet–Bloch sidebands. The non-trivial nature

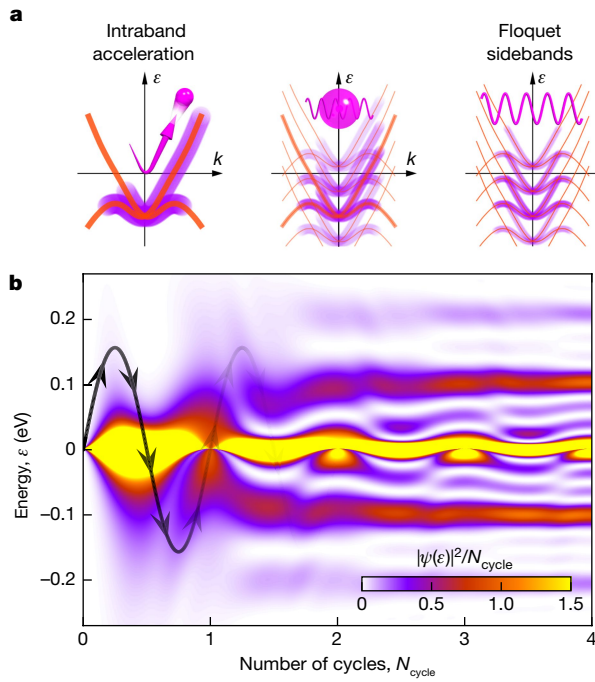
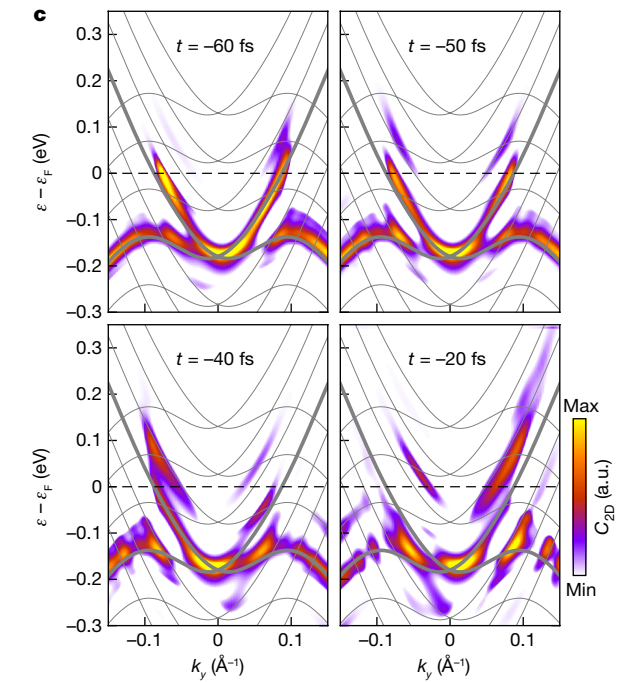


Fig. 3 | Lightwave ARPES calculations capturing the birth of Floquet–Bloch states. **a**, Illustration of light–matter interaction in the strong-field regime. Left: electron acceleration along a dispersing band as observed by a short probe pulse. Right: Floquet–Bloch states observed by continuous-wave probing. Under the present experimental conditions, both aspects simultaneously modulate the electronic distribution on the subcycle timescale (middle). **b**, Dependence of Floquet–Bloch energy spectra on the number of pump optical cycles, obtained from a minimal quantum model (Methods). The black

of this process is clearly visible, for example, from the fact that at $t = -20$ fs (Fig. 3c), the strongest weight does not reside in the equilibrium band as expected from the single-state picture of Fig. 3b, but is rather transferred almost entirely to the first sideband.

A Floquet path for interband transitions

Our experiments also shed light on a long-standing question in HHG, namely, how multiphoton excitation between equilibrium bands proceeds in the strong-field limit. In the three-dimensional topological insulator Bi_2Te_3 , the direct bulk bandgap at the Γ point exceeds 0.3 eV and thus prevents resonant one-photon interband transitions²⁸ ($\hbar\Omega_{\text{MIR}} = 0.1$ eV). A finite density of electrons can nonetheless dynamically bridge the gap following an intriguing quantum path through Floquet–Bloch replicas of the TSS. This is best seen in subcycle-ARPES maps (Fig. 4a) taken near and after the field crest of the MIR waveform. At the peak field of 0.8 MV cm^{-1} (Fig. 4b), the electronic distribution deviates markedly from the original band dispersion (Fig. 4a, $t = -12$ fs): a large fraction of the photoelectron signal is transferred to the first and second Floquet–Bloch sidebands, where the electrons are periodically slushed back and forth by the carrier field (Fig. 4a, $t = 22$ fs and 46 fs). The electron transfer changes critically on the scale of a few femtoseconds. As the experiment is performed under the conditions of HHG²⁸, our data capture electron dynamics underlying HHG directly in momentum space. Once the high-energy tail of the Floquet electron distribution reaches the bulk conduction band (BCB; Fig. 4a, grey area), the surface states hybridize with the bulk band and populate it. The corresponding photoelectron signature persists even after the MIR pulse is over (Fig. 4c), confirming transitions into the BCB.



curve indicates the trajectory of a massless Dirac electron (Fermi velocity $v_F = 4.1 \text{ Å fs}^{-1}$) in the periodic driving field. **c**, Lightwave ARPES maps calculated at four temporal steps of $t = -60$ fs, $t = -50$ fs, $t = -40$ fs and $t = -20$ fs. Curvature image filtering has been applied for consistency with the experimental data in Fig. 2c. See Methods for further details of the filtering procedure. Grey curves indicate the TSS band structure as derived from the $\mathbf{k}\cdot\mathbf{p}$ Hamiltonian (thick lines) and shifted by integer multiples of the driving frequency (thin lines).

The novel mechanism of transitions between equilibrium bands via Floquet–Bloch states is faithfully reproduced by quantum non-equilibrium calculations. Figure 4d shows simulated subcycle-ARPES spectra at $t = 22$ fs and $t = 46$ fs. As in the experiment (Fig. 4a), strong photoelectron intensities are generated in an energy–momentum region where the second and third Floquet–Bloch sidebands overlap with the bulk states, creating a persistent BCB population. For a more quantitative analysis, we trace the energy–momentum integral of the measured photoelectron intensity over the BCB region (Fig. 4c, dashed trapezoid), I_{BCB} , as a function of t (Fig. 4e). After a steep increase during the MIR pulse, I_{BCB} settles to a long-lived plateau, reflecting the incoherent electron population scattered into the BCB. Remarkably, the onset of I_{BCB} is superimposed with pronounced oscillations (Fig. 4e), which are fingerprints of the coherent coupling of the transiently dressed TSS with the BCB as qualitatively confirmed by calculations (Fig. 4f). The population of the Floquet surface states (red solid curve) strongly oscillates owing to inter- and intra-sideband excitations. When the dressed states reach the conduction band while the surface–bulk coupling is still coherent, the electron population can be transferred into and out of the bulk (grey curve) as observed experimentally. Ultimately, scattering creates an incoherent bulk population, which depletes the carrier density in the surface state compared with a situation without bulk–surface coupling (red dashed curve) and accelerates the collapse of the Floquet band structure while the MIR pulse ends. The transition thus results from a highly nonlinear strong-field process that has now become directly observable.

The ultrafast build-up and collapse of the dressed band structure open an unprecedented dynamical view of one of the essential conditions for Floquet–Bloch states: dephasing. For dressed states to form, electrons need to coherently follow the driving field and enable multipath

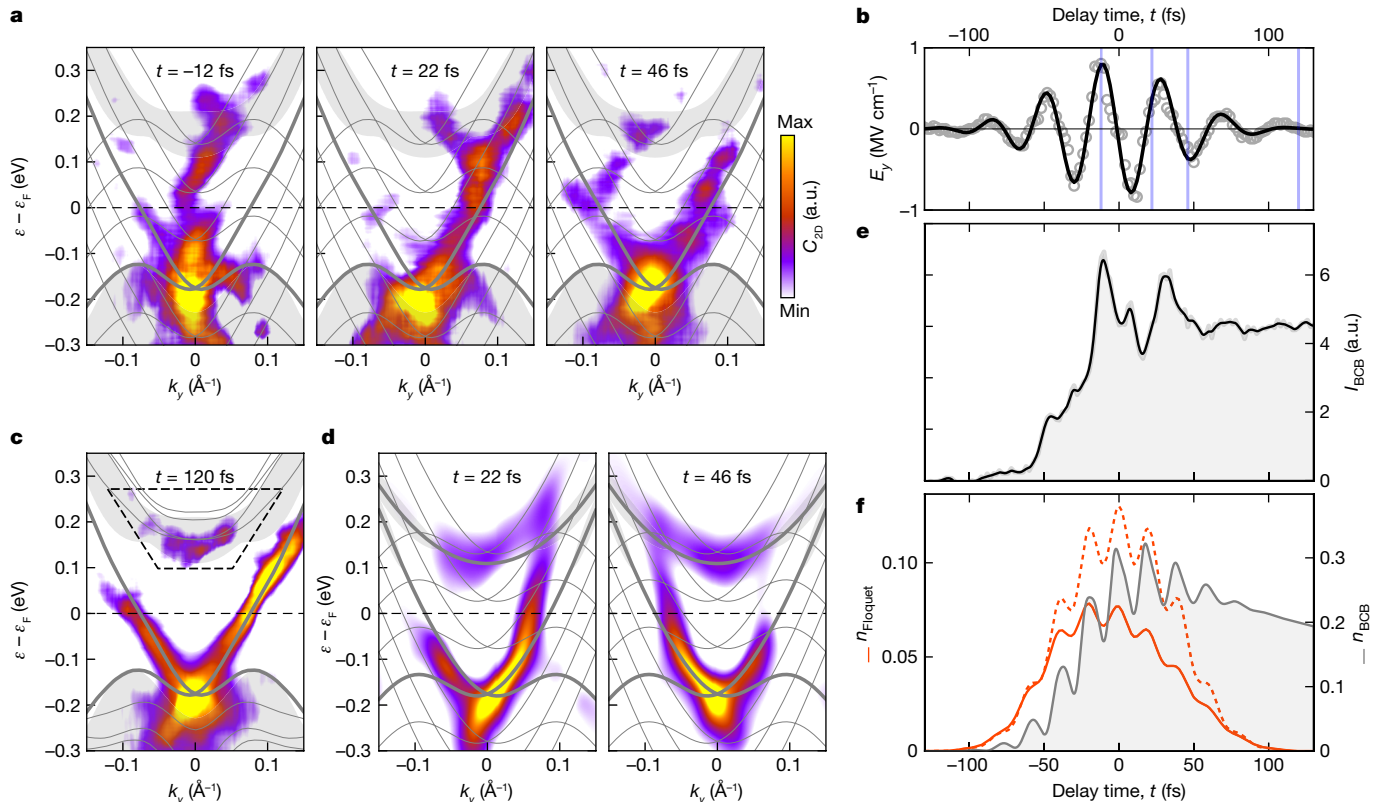


Fig. 4 | Interband transition to the BCB via dynamical Floquet–bulk coupling. **a**, Curvature-filtered lightwave ARPES maps as in Fig. 2c but measured at $t = -12$ fs, $t = 22$ fs and $t = 46$ fs. **b**, Experimental MIR driving waveform. The grey markers show the electric field from the streaking reconstruction and the black solid curve indicates its fit as in Fig. 2b. The thin vertical lines highlight the temporal steps in **a**. **c**, Measured ARPES map after MIR excitation ($t = 120$ fs). **d**, Curvature-filtered ARPES maps calculated at $t = 22$ fs and $t = 46$ fs with coupling of the TSS to bulk states. The thick grey curves describe the model band structure as in Fig. 3c. The grey shaded areas

illustrate delocalization of the bulk states to account for self-energy effects in our model (Methods). **e**, Temporal evolution of the experimental photoemission intensity from the BCB states obtained by integrating the raw ARPES data over the trapezoid shown in **c**. Grey curve and shading, experimental data; black curve, smoothed guide to the eye. **f**, Rise and decay of the population of Floquet surface states, n_{Floquet} , obtained from the calculated photoemission intensity with (red solid line) and without (red dashed line) coupling to bulk states. The grey curve describes the temporal evolution of the calculated occupation of the BCB, n_{BCB} .

quantum interference. From the surprisingly fast emergence of Floquet sidebands after only one oscillation cycle (Fig. 2c), we conclude that the quantum memory must survive at least one MIR oscillation period. This condition is exquisitely fulfilled for the TSS, where relaxation times exceeding 1 ps have been observed²⁴. Yet the bulk of Bi_2Te_3 forms a trivial insulator with relaxation times substantially shorter than $2\pi/\Omega_{\text{MIR}}$ (ref. 28). This explains why we observe Floquet–Bloch sidebands of the TSS, while there is no trace of analogous replicas of the bulk valence band (BVB) in the experimental ARPES maps (Figs. 2 and 4). Intriguingly, the electron transfer within the Floquet–Bloch bands dynamically changes coherence. When the dressed surface states can scatter into the bulk, they rapidly lose coherence, sealing the end of the dressed states. By increasing the driving frequency, we can boost this effect (Extended Data Fig. 10).

Conclusions and outlook

We truly visualized the birth, rise and collapse of Floquet–Bloch states in a topological insulator, in both experiment and quantum theory. By introducing subcycle ARPES in the MIR, we directly visualize in momentum space how the lightwave-driven surface state of Bi_2Te_3 crosses over from an initial regime of deterministic quantum trajectories to a Floquet band structure resulting from multipath quantum interference, within an Ehrenfest time³⁹ of one optical cycle. The dynamics unify the particle aspect of electrons describing subcycle acceleration with the

wave-like interference of electrons forming light-induced sidebands. Ultimately, electrons promoted in the emerging Floquet–Bloch band structure can scatter into the BCB, destroying the Floquet states. Our capability to image band structures that are transiently dressed and accelerated by strong light fields marks a milestone towards a complete understanding of Floquet engineering, HHG and other strong-field light–matter interaction concepts⁴⁰. Also, Floquet topological insulator phases in graphene^{1,9,14}, light-induced topological phase transitions in Weyl semimetals^{3,10} and dynamical Bloch oscillations triggering HHG bursts^{18–28} may now be visualized on the subcycle scale and, potentially, reveal ways to design quantum functionalities with light.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-023-05850-x>.

- Oka, T. & Aoki, T. Photovoltaic Hall effect in graphene. *Phys. Rev. B* **79**, 081406 (2009).
- Mitrano, M. et al. Possible light-induced superconductivity in K_3C_{60} at high temperature. *Nature* **530**, 461–464 (2016).
- Sie, E. J. et al. An ultrafast symmetry switch in a Weyl semimetal. *Nature* **565**, 61–66 (2019).
- Basov, D. N., Averitt, R. D. & Hsieh, D. Towards properties on demand in quantum materials. *Nat. Mater.* **16**, 1077–1088 (2017).

5. de la Torre, A. et al. Colloquium: Nonthermal pathways to ultrafast control in quantum materials. *Rev. Mod. Phys.* **93**, 041002 (2021).
6. Lindner, N. H., Refael, G. & Galitski, V. Floquet topological insulator in semiconductor quantum wells. *Nat. Phys.* **7**, 490–495 (2011).
7. Wang, Y. H., Steinberg, H., Jarillo-Herrero, P. & Gedik, N. Observation of Floquet–Bloch states on the surface of a topological insulator. *Science* **342**, 453–457 (2013).
8. Mahmood, F. et al. Selective scattering between Floquet–Bloch and Volkov states in a topological insulator. *Nat. Phys.* **12**, 306–310 (2016).
9. Sentef, M. A. et al. Theory of Floquet band formation and local pseudospin textures in pump–probe photoemission of graphene. *Nat. Commun.* **6**, 7047 (2015).
10. Hübener, H., Sentef, M. A., Giovannini, U. D., Kemper, A. F. & Rubio, A. Creating stable Floquet–Weyl semimetals by laser-driving of 3D Dirac materials. *Nat. Commun.* **8**, 13940 (2017).
11. Reutzel, M., Li, A., W. Z. & Petek, H. Coherent multidimensional photoelectron spectroscopy of ultrafast quasiparticle dressing by light. *Nat. Commun.* **11**, 2230 (2020).
12. Zhang, J. et al. Observation of a discrete time crystal. *Nature* **543**, 217–220 (2017).
13. Choi, S. et al. Observation of discrete time crystalline order in a disordered dipolar many-body system. *Nature* **543**, 221–225 (2017).
14. McIver, J. W. et al. Light-induced anomalous Hall effect in graphene. *Nat. Phys.* **16**, 38–41 (2020).
15. Träger, N. et al. Real-space observation of magnon interaction with driven space-time crystals. *Phys. Rev. Lett.* **126**, 057201 (2021).
16. Ghimire, S. et al. Observation of high-order harmonic generation in a bulk crystal. *Nat. Phys.* **7**, 138–141 (2011).
17. Zaks, B., Liu, R. B. & Sherwin, M. S. Experimental observation of electron–hole recollisions. *Nature* **483**, 580–583 (2012).
18. Schubert, O. et al. Sub-cycle control of terahertz high-harmonic generation by dynamical Bloch oscillations. *Nat. Photon.* **8**, 119–123 (2014).
19. Luu, T. T. et al. Extreme ultraviolet high-harmonic spectroscopy of solids. *Nature* **521**, 498–502 (2015).
20. Langer, F. et al. Lightwave-driven quasiparticle collisions on a subcycle timescale. *Nature* **533**, 225–229 (2016).
21. Vampa, G. et al. Linking high harmonics from gases and solids. *Nature* **522**, 462–464 (2015).
22. Yoshikawa, N., Tamaya, T. & Tanaka, K. High-harmonic generation in graphene enhanced by elliptically polarized light excitation. *Science* **356**, 736–738 (2017).
23. Higuchi, T., Heide, C., Ullmann, K., Weber, H. B. & Hommelhoff, P. Light-field-driven currents in graphene. *Nature* **550**, 224–228 (2017).
24. Reimann, J. et al. Subcycle observation of lightwave-driven Dirac currents in a topological surface band. *Nature* **562**, 396–400 (2018).
25. Hafez, H. et al. Extremely efficient terahertz high-harmonic generation in graphene by hot Dirac fermions. *Nature* **561**, 507–511 (2018).
26. Langer, F. et al. Lightwave valleytronics in a monolayer of tungsten diselenide. *Nature* **557**, 76–80 (2018).
27. Borsch, M. et al. Super-resolution lightwave tomography of electronic bands in quantum materials. *Science* **370**, 1204–1207 (2020).
28. Schmid, C. P. et al. Tunable non-integer high-harmonic generation in a topological insulator. *Nature* **593**, 385–390 (2021).
29. Cheng, B. et al. Efficient terahertz harmonic generation with coherent acceleration of electrons in the Dirac semimetal Cd_3As_2 . *Phys. Rev. Lett.* **124**, 117402 (2020).
30. Hohenleutner, M. et al. Real-time observation of interfering crystal electrons in high-harmonic generation. *Nature* **523**, 572–575 (2015).
31. Schultze, M. et al. Attosecond band-gap dynamics in silicon. *Science* **346**, 1348–1352 (2014).
32. Hasan, M. Z. & Kane, C. L. Colloquium: Topological insulators. *Rev. Mod. Phys.* **82**, 3045 (2010).
33. Chen, Y. L. et al. Experimental realization of a three-dimensional topological insulator, Bi_2Te_3 . *Science* **325**, 178–181 (2009).
34. Holthaus, M. Floquet engineering with quasienergy bands of periodically driven optical lattices. *J. Phys. B* **49**, 013001 (2016).
35. Ikeda, T. N., Tanaka, S. & Kayanuma, Y. Floquet–Landau–Zener interferometry: usefulness of the Floquet theory in pulse-laser-driven systems. *Phys. Rev. Res.* **4**, 033075 (2022).
36. Aeschlimann, S. et al. Survival of Floquet–Bloch states in the presence of scattering. *Nano Lett.* **21**, 5028–5035 (2021).
37. Zhang, P. et al. Observation of topological superconductivity on the surface of an iron-based superconductor. *Science* **360**, 182–186 (2018).
38. Maja-Avila, L. et al. Laser-assisted photoelectric effect from surfaces. *Phys. Rev. Lett.* **97**, 113604 (2006).
39. Maldacena, J. et al. A bound on chaos. *J. High Energy Phys.* **2016**, 106 (2016).
40. Ikeda, T. N. & Sato, M. General description for nonequilibrium steady states in periodically driven dissipative quantum systems. *Sci. Adv.* **6**, eabb4019 (2020).

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